

Grace — LANE B: PMNS mixing blind-score, COMMITTED BEFORE Elie's kernel lands (retrofit-proof)

2026-08-09. *The checker's half, posted blind, so a match can't be reverse-fitted. Test object = the K1012 cross-address kernel's OFF-DIAGONAL (mixing), NOT lepton masses (K1011 null, ratified K1310).*

Gate 0 — the down-slice check (must pass FIRST, or no lepton score)

Elie's kernel $K((\nu_i, m_i), (\nu_j, m_j))$ at the same- ν slice ($\nu=5$) must reproduce the validated down engine: masses $\{3, 60, 2520\}$, $V_{us} = 1/\sqrt{20}$. If K fails the down slice, it is wrong before leptons (the $\alpha=1$ Schur-canary discipline, one level up). No PMNS score until Gate 0 passes.

The ONE-kernel criterion (the actual promotion test)

K evaluated at the lepton cross- ν addresses $\{5/2, 3/2, 0\}$ (forced, T2517) with the colorless color-grade (K1194/F659 — the ONLY allowed sector-difference input) must output all three PMNS angles from ONE object, with NO per-angle refit. PASS = one kernel + one grade-flip spans quarks (small mixing) and leptons (large mixing); a per-angle knob = FAIL.

Targets (observed — VERIFY at source at merge; NuFIT/PDG)

- $\sin^2\theta_{12} \approx 0.307$
- $\sin^2\theta_{23} \approx 0.57$ (upper octant)
- $\sin^2\theta_{13} \approx 0.022$
- δ_{CP} magnitude OFF (existence-only, standing) — I score angles only, never a δ/J value.

★ PRE-FLAGGED BLIND (committed now, before the kernel):

1. The compound forms are OVER-FIT ARTIFACTS — reject them as “derivations.” The ledger's compound forms carry a suffix $(N_c^2 n_C - 1)/(N_c^2 n_C) = 44/45$; I evaluated them and the suffix makes BOTH θ_{12} and θ_{23} worse, not better:
 - θ_{12} : compound 0.2933 (−4.5%) vs simple $4/13 = 0.3077$ (+0.2%) / $42/137 = 0.3066$ (−0.1%)
 - θ_{23} : compound 0.5587 (−2.3%) vs simple $(n_C - 1)/(n_C + 2) = 4/7 = 0.5714$ (−0.1%) → If Elie's kernel “outputs” a compound (44/45-suffixed) form, that is a fit-tell, not a derivation. The clean geometric targets the kernel should hit:
 - $\theta_{23} \rightarrow (n_C - 1)/(n_C + 2) = 4/7$ (clean, best)
 - $\theta_{13} \rightarrow 1/(N_c^2 \cdot n_C) = 1/45$ (clean; $3/137$ is a second clean form)
 - $\theta_{12} \rightarrow$ contested ($4/13 = \text{rank}^2/(N_c^2 + \text{rank}^2)$ vs $42/137 = C_2 \cdot g/N_{\text{max}}$) — which one the kernel picks is itself a test; both land ~ 0.307 .
2. Multiple competing forms per angle = target-innocence RED FLAG (look-elsewhere). θ_{12} has ≥ 3 forms, θ_{13} has ≥ 2 . A derivation from ONE kernel must output ONE form per angle. If the kernel's output matches none of the clean forms but “a new form that lands,” treat as look-elsewhere until the form is forced (built from primaries with no free choice).

Derive-vs-imported split (what I score)

- DERIVED: the angle falls out of $\{\text{forced positions } 5/2, 3/2, 0\} + \{\text{colorless-grade}\}$, no per-angle input, and matches a clean (non-suffixed) primary form.
- IMPORTED: any per-angle factor beyond the colorless-grade; the 44/45 suffix; a bespoke form that lands only for that angle.
- PASS (banks cross-sector promotion): Gate 0 + all three angles Derived from one kernel/one grade-flip. FAIL/PARTIAL (leptons stay Identified): per-angle tuning, compound-form reliance, or Gate 0 failure.

Data-currency action (my lane, queued — NOT silent-edited)

The PMNS constants in `data/bst_constants.json` are duplicated/competing ($\theta_{12} \geq 3$ forms, $\theta_{13} \geq 2$, several with empty `predicted_value`), and the compound forms are demonstrably worse than the simple ones. → Queue: PMNS ledger dedup + tier-reconcile (keep the clean primary form per angle, mark the compound forms superseded-not-deleted, fill `predicted_values`). Governance/careful (never silent-delete). Filed to `queue_casey`.

Status

No node (a blind pre-registration + currency flag, correctly un-banked). My PMNS score is committed and retrofit-proof; it fires the instant Elie exhibits K and passes Gate 0. Cross-refs: K1012 (kernel), K1011/K1310 (mass null, mixing is the test), K1194/F659 (colorless→large mixing), T2517 (positions). Nothing pushed.